

Part I

SUMMARY of Product Rule and Quotient Rule

Summary

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Summary

- **The Product Rule:** If f and g are differentiable functions, then $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$.
- **The Quotient Rule:** If f and g are differentiable functions, then $\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$.

Part II

Recitation Questions

Problem 1

February 21, 2026

Problem 1

Problem 1 *Differentiate the functions using product or quotient rule.*

(a) $h(u) = 7ue^u$

(b) $s(t) = \frac{\sqrt{t}}{e^t}$

Problem 2

February 21, 2026

Problem 2

Problem 2 *Differentiate the function f defined by $f(x) = 1/x^8$ in two different ways.*

Problem 3

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Problem 3

Problem 3 Suppose that $f(5) = 7$, $f'(5) = 8$, $g(5) = 3$, and $g'(5) = -4$. Find:

(a) $(fg)'(5)$.

(b) $\left[\frac{d}{dx} \left(\frac{f}{g} \right) \right]_{x=5}$

(c) $\left(\frac{g}{f} \right)'(5)$

(d) $\left[\frac{d}{dx} \left(\frac{g(x)}{x+2} \right) \right]_{x=5}$

(e) $\left[\frac{d}{dx} \left(\frac{xf(x)}{g(x)} \right) \right]_{x=5}$

(f) $\lim_{x \rightarrow 5} \frac{x^2 g(x) - 75}{x - 5}$. **EXPLAIN.**

(g) $\lim_{x \rightarrow 5} \frac{\frac{x+2}{f(x)} - 1}{x - 5}$. **EXPLAIN.**

Problem 4

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Problem 4

Problem 4 Use the given information to find the equation of the tangent line.

(a) Given $g(x) = x^3 f(x)$, $f(2) = 4$, and $f'(2) = 7$, find the equation of the tangent line to the graph of g at $x = 2$.

(b) Given $h(z) = \frac{zs(z)}{z-3}$, $s(2) = 4$, and $s'(2) = 7$, find the equation of the tangent line to the graph of h at $z = 2$.

(c) Given

x	1	2	3	4	5
$f(x)$	5	3	0	-4	3
$f'(x)$	-3	-5	-2	6	-4
$g(x)$	6	9	-8	13	15
$g'(x)$	8	5	-10	7	6

find the equation of the tangent line of

$$\frac{f(x)}{e^x g(x)}$$

at $x = 2$.

Problem 5

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Problem 5

Problem 5 Differentiate the following functions:

(a) Given $f(x) = (x^2 + 4x - 7)e^{-x}$, show $f'(x) = (11 - x^2 - 2x)e^{-x}$.

(b) Given $g(x) = \frac{x^2 + 4x - 7}{e^{-x}}$, show $g'(x) = \frac{x^2 + 6x - 3}{e^{-x}}$

Problem 6

February 21, 2026

Problem 6

Problem 6 The graph of a function g is given below. Using the graph, estimate the derivative at the given point:

$$\frac{d}{dx} \left(\frac{xg(x)}{x+3} \right) \text{ at } x = 1$$

